

SQL Date Functions

Built-in tools used to manipulate, format, calculate, and extract values from columns containing dates and timestamps.

SQL DATE FUNCTIONS

Part Extraction

DAY
MONTH
YEAR
DATEPART
DATENAME
DATETRUNC
EOMONTH

Format & Casting

FORMAT
CONVERT
CAST

Calculations

DATEADD
DATEDIFF

Validation

ISDATE

PRACTICE ENVIRONMENT TESTING SETUP (MYDB.LOGISTICS)

```
USE mydb;
CREATE TABLE Logistics (
  ShipmentID INT,
  Location VARCHAR(50),
  Ship_Date DATETIME
);

INSERT INTO Logistics VALUES
(101, 'Mumbai', '2026-06-15 08:30:00'), (102, 'Pune', '2026-06-16 10:15:00'),
(103, 'Delhi', '2026-06-17 14:45:00'), (104, 'Chennai', '2026-06-18 09:20:00'),
(105, 'Kolkata', '2026-06-19 16:10:00'), (106, 'Bengaluru', '2026-06-20 11:55:00');
```

Part Extraction

Isolating a specific component of a date or timestamp—such as just the Year, Month, or Day—and pulling it out as an independent value.

Real-World Use Cases: **Trend Analysis:** Grouping registrations by month to analyze growth patterns. | **Behavioral Patterns:** Extracting weekdays to monitor traffic variance. | **Seasonality:** Isolating hours to locate server peak loads.

DAY()

DAY(date)

Extracts the day component of the specified date as an integer value.

```
SELECT Location, Ship_Date, DAY(Ship_Date) AS day_ FROM Logistics;
```

Sample Output Format: Mumbai | 15-06-2026 08:30 | 15

MONTH()

MONTH(date)

Returns an integer representing the calendar month numeric index (1-12).

```
SELECT Location, Ship_Date, MONTH(Ship_Date) AS month_ FROM Logistics;
```

Sample Output Format: Mumbai | 15-06-2026 08:30 | 6

YEAR()

YEAR(date)

Isolates and extracts the isolated 4-digit calendar year value.

```
SELECT Location, Ship_Date, YEAR(Ship_Date) AS year_ FROM Logistics;
```

Sample Output Format: Mumbai | 2026-06-15 08:30:00 | 2026

DATEPART()

DATEPART(part, date)

Returns a precise user-defined date part component output as a calculated integer identifier.

```
SELECT ShipmentID, DATEPART(YEAR, Ship_Date) AS Year, DATEPART(MONTH, Ship_Date) AS MONTH FROM Logistics;
```

Sample Output Format: 101 | 2026 | 6

DATENAME()

DATENAME(part, date)

Returns a descriptive text string character layout pattern representing the specified target component.

```
SELECT ShipmentID, DATENAME(MONTH, Ship_Date) AS MONTH_Txt, DATENAME(WEEKDAY, Ship_Date) AS WeekDay_Txt FROM Logistics;
```

Sample Output Format: 101 | June | Monday

DATETRUNC()

DATETRUNC(part, date)

Truncates a date expression down to a specified alignment unit, resetting smaller elements to zero baselines.

```
SELECT DATETRUNC(MONTH, '2026-06-18 09:20:35.123');
```

Sample Output Format: 2026-06-01 00:00:00.000

EOMONTH()

EOMONTH(date, [offset])

Finds and returns the absolute last calendar day of the month for the date expression with an optional offset.

```
SELECT EOMONTH('2026-06-18', 1);
```

Sample Output Format: 2026-07-31

Format & Casting

Converts date data types into alternative format profiles or translates layout views for system reporting structures.

FORMAT() FORMAT(date, format_pattern)

Formats a date expression into a custom localized human-readable character layout pattern string.

```
SELECT FORMAT(Ship_Date, 'dd-MM-yyyy') AS FormattedDate FROM Logistics;
```

Sample Output Format: 15-06-2026

```
SELECT FORMAT('2026-06-18', 'MMMM') AS MonthName;
```

Sample Output Format: June

CAST() / CONVERT() CAST(expr AS type)

Translates data types directly into standardized date or time schema views.

```
SELECT CAST(Ship_Date AS DATE) AS DispatchDate FROM Logistics;
```

Sample Output Format: 2026-06-15

Calculations

Executes interval addition/subtraction or gauges explicit positional chronological distance values between endpoints.

DATEADD() DATEADD(part, increment, date)

Appends or subtracts a specific temporal step increment value directly to a target validation timestamp.

```
SELECT ShipmentID, DATEADD(DAY, 7, Ship_Date) AS Plus7Days FROM Logistics;
```

Sample Output Format: 101 | 2026-06-22 08:30:00

DATEDIFF() DATEDIFF(part, start_date, end_date)

Measures and returns the total sequence boundary interval gaps crossed between two chronological markers.

```
SELECT ShipmentID, DATEDIFF(DAY, Ship_Date, GETDATE()) AS DaysSinceDispatch FROM Logistics;
```

Sample Output Format: 101 | 5

Validation

Evaluates data integrity to confirm that strings safely parse as correct structural calendar entries.

ISDATE() ISDATE(expression)

Verifies if an input expression forms a valid datatype format, returning 1 (true) or 0 (false).

```
SELECT ISDATE(Ship_Date) AS Result FROM Logistics;
```

Sample Output Format: 1